

Calculus & Analytical Geometry

Assignment 1

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BCS-3Z

Solutions:

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Question 1.

a) $\lim_{x \rightarrow -7} f(x)$

$\lim_{x \rightarrow -7} f(x)$ gives $y=0$ but x approaches $-\infty$.

b) $\lim_{x \rightarrow -3} f(x)$

$\lim_{x \rightarrow -3} f(x)$ does not exist as $\lim_{x \rightarrow -3^-} = -\infty$ while $\lim_{x \rightarrow -3^+} = +\infty$.

Hence left hand limit isn't equal to right hand limit.

c) $\lim_{x \rightarrow 0} f(x)$

$\lim_{x \rightarrow 0} f(x) = 0$

d) $\lim_{x \rightarrow 6^-} f(x)$

$\lim_{x \rightarrow 6^-} f(x) = 1$

e) $\lim_{x \rightarrow 6^+} f(x)$

$\lim_{x \rightarrow 6^+} f(x) = 1$

f) Equations of vertical asymptotes

$\lim_{x \rightarrow -3} f(x) = \frac{1}{x+3}$

$\lim_{x \rightarrow 2} f(x) = -\frac{1}{(x-2)^2}$

$$\lim_{x \rightarrow 5} f(x) = \frac{1}{(x-5)^2}$$

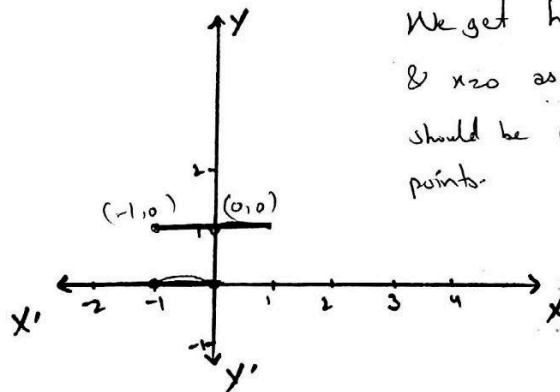
Question 2:

Domain of $f = [-1, 1]$

$$f(-1) = f(0) = 0$$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 1} f(x) = 1$$

$$f(x) = \begin{cases} 0 & , x = -1, x = 0 \\ 1 & , -1 < x < 0 \text{ or } 0 < x \leq 1 \end{cases}$$



We get holes at $x = -1$ & $x = 0$ as the value of y should be 0 at these 2 points.

Question No 3

$$\lim_{t \rightarrow 9} \frac{9-t}{53-t}$$

$$\lim_{t \rightarrow 9} f(t) = \lim_{t \rightarrow 9} \frac{9-t}{53-t}$$

~~Rationalizing the $f(t)$~~

$$\lim_{t \rightarrow 9} (f(t)) = \lim_{t \rightarrow 9} \frac{(9-t)(53+t)}{3-t^2}$$

$$\lim_{t \rightarrow 9} \frac{9-t}{53-t} = \lim_{t \rightarrow 9} \frac{9-t}{53-t}$$

$$= \frac{9-9}{53-9}$$

$$= \frac{0}{53-9} \Rightarrow 0$$

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$$ii - \lim_{y \rightarrow -\infty} \frac{2-y}{\sqrt{7+6y^2}}$$

$$\lim_{y \rightarrow -\infty} \frac{2-y}{\sqrt{7+6y^2}} = \lim_{y \rightarrow -\infty} \frac{-y(\frac{2}{y} - 1)}{\sqrt{-y^2(\frac{7}{-y^2} + 6)}}$$

$$= \lim_{y \rightarrow -\infty} \frac{-y(\frac{2}{y} - 1)}{-y\sqrt{\frac{7}{y^2} + 6}}$$

$$= \lim_{y \rightarrow -\infty} \frac{\frac{2}{y} - 1}{\sqrt{\frac{7}{y^2} + 6}}$$

$$= \frac{0-1}{\sqrt{0+6}} \Rightarrow -\frac{1}{\sqrt{6}}$$

$$iii - \lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{e^x}{e^x} + \frac{e^{-x}}{e^x}}{\frac{e^x}{e^x} - \frac{e^{-x}}{e^x}} \Rightarrow \lim_{x \rightarrow \infty} \frac{1 + e^{-2x}}{1 - e^{-2x}}$$

$$= \frac{1+0}{1-0} \Rightarrow 1$$

$$iv - \lim_{x \rightarrow \pi} \cos^2(x - \tan x)$$

$$= \lim_{x \rightarrow \pi} \cos^2(x - \tan x) \Rightarrow \cos^2(\pi - \tan \pi)$$

$$= \cos^2(\pi - 0) \Rightarrow \cos^2(\pi)$$

$$= (-1)^2 \Rightarrow 1$$

Question No 4.

$$\lim_{x \rightarrow 0} (x^2 \cos 2\pi x) = 0$$

$$-1 < \cos 2\pi x < 1$$

Multiplying x^2 on both sides

$$-x^2 < x^2 \cos 2\pi x < x^2$$

$$f(x) < g(x) < h(x)$$

$$\lim_{x \rightarrow 0} (-x^2) = 0 \quad ; \quad \lim_{x \rightarrow 0} (x^2) = 0$$

Hence, $x^2 \cos 2\pi x$ is also '0'.

Question No 5.

$$\frac{f(x)}{g(x)} \quad ; \quad g(x) \neq 0$$

Vertical asymptotes are not possible, however, horizontal & oblique asymptotes are still possible if $\lim_{x \rightarrow \infty} f(x)$ and degree of $f(x)$ is less than degree of $g(x)$ respectively.

Vertical asymptotes are not possible because $g(x)$ can not be equal to zero anywhere.

Question No 6.

$$f(-2)=3, \quad f(-1)=-1, \quad f(0)=-4, \quad f(1)=1, \quad f(2)=5$$

$$[-2, -1]$$

$$f(-2)=3, \quad f(-1)=-1$$

$$f(-2) = 3 > 0 \text{ while } f(-1) = -1 < 0$$

Hence a root exists in $[-2, -1]$.

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b. $[-1, 0]$

$$f(-1) = -1 < 0$$

$$f(0) = -4 < 0$$

• Since both are negative, no roots exist as per Intermediate value theorem.

c. $[0, 2]$

$$f(0) = -4 < 0$$

$$f(2) = 5 > 0$$

• Hence root exists in this interval.

Question 7:-

$$x^2 + 5 \quad x > 2$$

$$m(x+1) + k \quad -1 < x \leq 2$$

$$2x^3 + x + 7 \quad x \leq -1$$

$$\lim_{x \rightarrow -1} f(x) = 2(-1)^3 + (-1) + 7$$

$$= -2 - 1 + 7 = 4$$

$$\lim_{x \rightarrow 2^-} f(x) = m(3) + k$$

$$= 3m + k$$

$$\lim_{x \rightarrow 2^+} f(x) = 4 + 5 = 9$$

$$3m + k = 9 \quad (\text{Considering L.H.L \& R.H.L equal})$$

$$3m = 9 - k$$

$$m = \frac{9 - k}{3}$$

$$\text{Using } x \rightarrow -1^+$$

$$\lim_{x \rightarrow -1^+} f(x) = m(0) + k = k$$

$$k = 4$$

$$\text{Put } k = 4 \text{ in } m = \frac{9 - k}{3}$$

$$m = \frac{9 - 4}{3} = \frac{5}{3}$$

If $m = \frac{5}{3}$ & $k = 4$, the $f(x)$ will be continuous.

Question 8.

$$i - \frac{2x^3 + 7}{x^3 - x^2 + x + 7}$$

For vertical asymptotes

$$x^3 - x^2 + x + 7 = 0$$

$$x^3 - x^2 + x = -7$$

Hit & Trial method.

$$\text{Put } x = 1$$

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$$1 - 1 + 1 = -7$$

$$1 \neq -7$$

$$x = 2$$

$$8 - 4 + 2 = -7$$

$$6 \neq -7$$

No ~~real~~ ^{rational} roots in this regard.

Try with $x = -1.5$

$$(-1.5)^3 + (-1.5)^2 + (-1.5) = -7$$

$$\frac{-23}{8} \approx -2.875 \quad r1.488 \approx -1.5$$

$$\frac{-23}{8} \approx -2.875$$

$\Rightarrow x = -1.5$ gives the vertical asymptote.

• For Horizontal Asymptotes, put $x \rightarrow \infty$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^3 \left(2 + \frac{7}{x^3}\right)}{x^3 \left(1 - \frac{x^2}{x^3} + \frac{11}{x^3} + \frac{7}{x^3}\right)}$$

$$= \frac{2+0}{1-0} \Rightarrow 2$$

$\Rightarrow f(x) = 2$ gives horizontal asymptote.

• Since degree of numerator and denominator is same,
and horizontal asymptote exists, so oblique asymptote doesn't exist.

Question 10:

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$$f(x) = \sqrt[3]{|9-x^2| - 1}$$

$$f(x) = \sqrt[3]{|9-x^2|} - 1$$

$$\Rightarrow \text{Domain} = \mathbb{R}$$

$$\Rightarrow \text{Range} = [-1, \infty)$$

$$\text{Zeros} = f(x) = 0$$

$$\sqrt[3]{|9-x^2|} = 1$$

$$|9-x^2| = 1$$

$$\Rightarrow x^2 = 8, \quad x^2 = 10$$

$$\Rightarrow x = \sqrt{8}, \quad x = \sqrt{10}$$

Question 11:

$$x^2 - 12x + 3y = 1$$

Since the equation is reflected by the origin, put $x = -x$ & $y = -y$.

$$(-x)^2 - 12(-x) + 3(-y) = 1$$

$$x^2 + 12x - 3y = 1$$

$$f(x) = \frac{x^2 + 12x - 1}{3}$$

ii- $\frac{3x^3 + 5x^2 - 1}{6x^3 - 7x + 3}$

For vertical asymptotes

$$6x^3 - 7x + 3 = 0$$

By hit and trial method

$\Rightarrow x \approx 1.21$ gives vertical asymptote

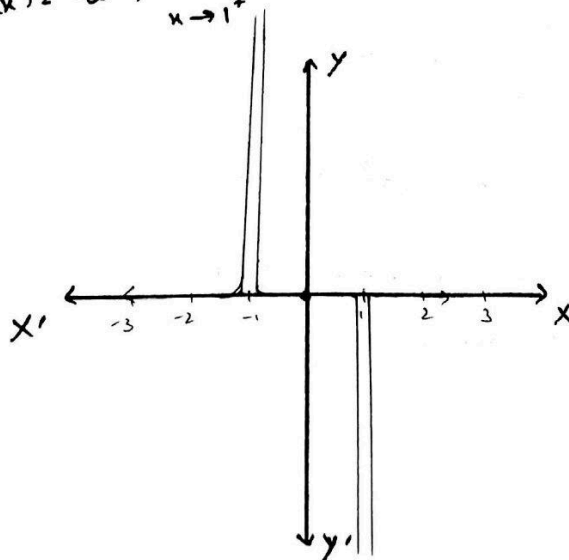
- Horizontal Asymptote doesn't exist as horizontal asymptote requires equal degree of numerator & denominator.
- Oblique Asymptote doesn't exist as there's difference of more than one degree among numerator & denominator.

Question 9.

$$f(0) = 0, \quad \lim_{x \rightarrow \pm\infty} f(x) = 0$$

$$\lim_{x \rightarrow -1^-} f(x) = \infty, \quad \lim_{x \rightarrow -1^+} f(x) = \infty$$

$$\lim_{x \rightarrow 1^-} f(x) = -\infty, \quad \lim_{x \rightarrow 1^+} f(x) = -\infty$$



Question 12.

$$f(x) = 5x$$

Since $f(x)$ is reflected across y-axis,

$$g(x) = 5x$$

After stretch by factor of '2', $g(x)$ becomes:

$$\Rightarrow g(x) = 2 \cdot 5x$$

$$\begin{aligned} \text{Domain } g(x) &= -x \geq 0, \quad x \leq 0 \\ &= (-\infty, 0]. \end{aligned}$$

Question No 13.

$$f(x) = \frac{2x+1}{x-3}$$

$\Rightarrow x-3 \geq 0 \Rightarrow x \geq 3$ gives vertical asymptote.

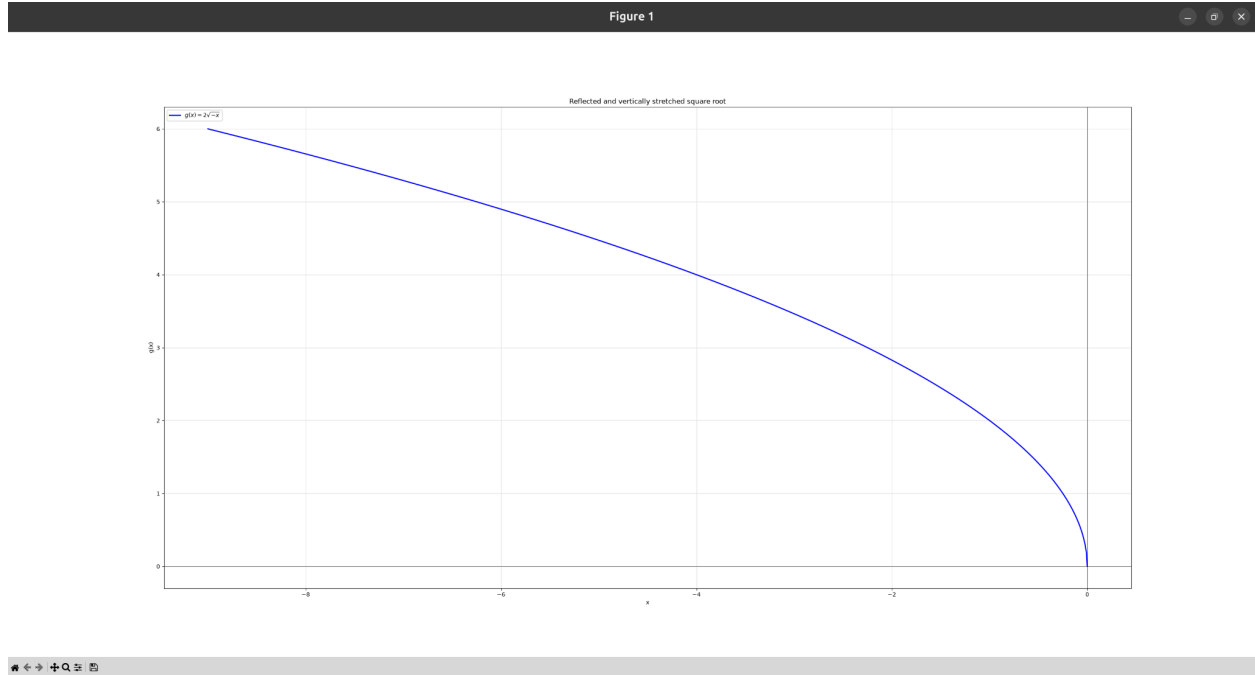
$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x(2 + 1/x)}{x(1 - 3/x)}$$

$$\lim_{x \rightarrow \infty} f(x) = \frac{2}{1} \Rightarrow 2$$

• No oblique asymptote as deg of num = deg. of denom.

Python Verification:

Q12:



Q13:

```
File Edit Selection View Go Run ... Sem3
Chat
> Ports
Terminal
(.venv) muhammad-awais@ubuntu:~/Work/University-Assignments/Sem3/Calculus/Assignment1$ python3 verifyQ13.py
Limit as x -> 3+: 00
Limit as x -> 3-: -00
Limit as x -> +00: 2
Limit as x -> -00: 2
Degree numerator: 1
Degree denominator: 1
o (.venv) muhammad-awais@ubuntu:~/Work/University-Assignments/Sem3/Calculus/Assignment1$
```

bash - Assignment1

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